

3 Whales are large mammals.

The photograph shows a beluga whale and a human diver.



(Source: © Andrey Nekrasov/Alamy Stock Photo)

(a) (i) A beluga whale can become habituated to human divers.

How many of the following statements about habituation are correct?

(1)

- habituation is an example of anatomical adaptation
- habituation is only observed in mammals
- habituation only occurs after repeated exposure to the same stimulus

☐ **A** 0

☐ **B** 1

☐ **C** 2

☐ **D** 3

(ii) A beluga whale can respond to a stimulus using a spinal reflex arc.

Which row is correct?

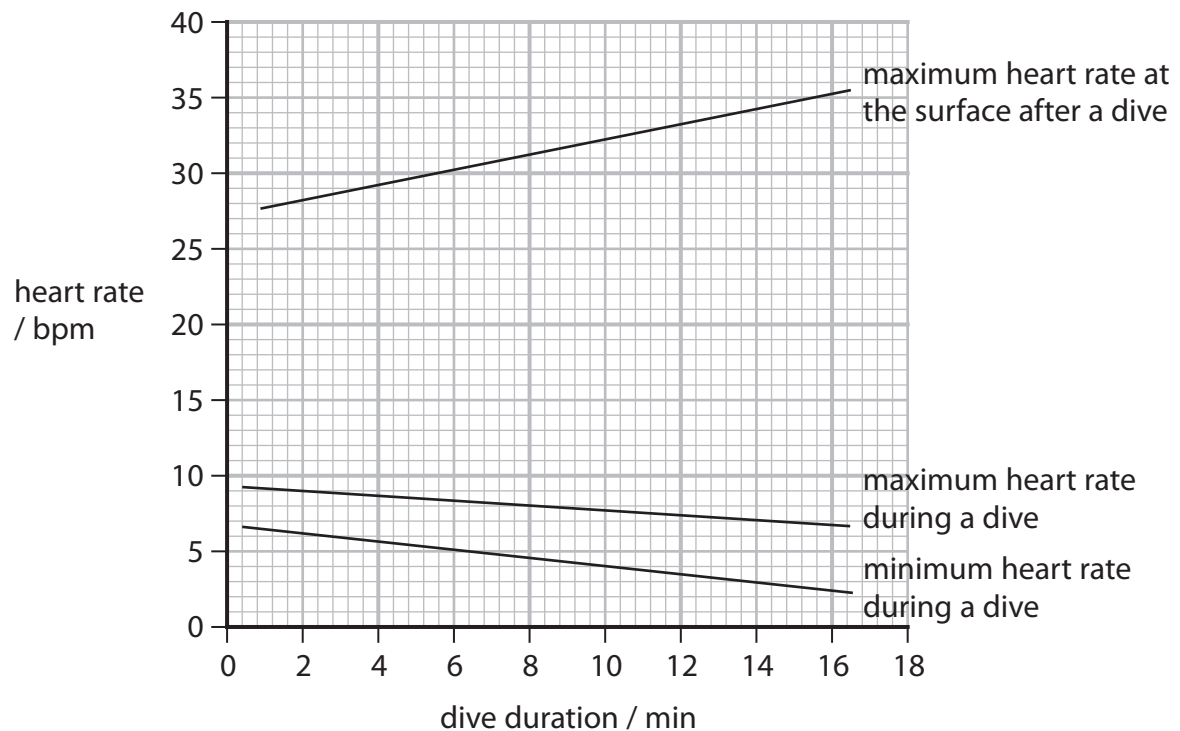
(1)

	Location of relay neurone	Location of cell body on the sensory neurone
☐ A	grey matter of spinal cord	at the end of the axon
☐ B	grey matter of spinal cord	in the middle of the axon
☐ C	white matter of spinal cord	at the end of the axon
☐ D	white matter of spinal cord	in the middle of the axon

(b) The heart rate of a whale changes when it is diving.

The graph shows the changes in heart rate that occur during dives of different durations.

The maximum heart rate at the surface is when the whale returns to the surface after a dive.



(i) Comment on the changes in heart rate of the whale when diving.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Explain how the changes in heart rate are controlled when the whale returns to the surface immediately after the dive.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 3 = 8 marks)